Customer Churn & Lifetime Value (LTV) Engine

Production Machine Learning, LTV Forecasting, FastAPI Microservice & Executive BI Suite

Author: Adarsh Kumar Pandey	**Dataset:** IBM Watson Telco Cohort (7,043 Records)
GitHub: https://github.com/addu91577-sketch/Customer-Churn-LTV-Engine	**Stack:** Python, FastAPI, XGBoost, Docker, PostgreSQL, Power BI

Executive Overview & High-Impact KPIs

Customer attrition in telecommunications incurs up to **5x higher replacement costs** than proactive subscriber retention. This project presents an enterprise-grade analytics engineering and predictive intelligence lifecycle that ingests raw telemetry into a PostgreSQL data warehouse via SQLAlchemy, executes granular exploratory data analysis (EDA), benchmarks ensemble classification models (XGBoost, Random Forest, Logistic Regression), unpacks feature importance via game-theoretic SHAP explainability, forecasts active subscriber Lifetime Value ($R^2 = 0.9983$), and deploys real-time FastAPI endpoints containerized in Docker alongside multi-tier Power BI and Metabase executive dashboards.

Total Base	Churn Rate	Lost MRR	Annual Risk	Avg LTV	LTV Model R^2
7,043	**26.54%**	**$139.13K**	**$2.45M**	**$3,310**	**0.9983**
Audited Cohort	1,869 Lost Users	Monthly Revenue Drag	ML Risk Exposure	Subscriber Pool	Predictive Accuracy

1. Data Ingestion & Storage: PostgreSQL & SQLAlchemy Engine

A robust relational foundation was engineered to transform raw CSV flat files into an enterprise-grade schema within PostgreSQL, enabling structured relational queries, ACID guarantees, and direct downstream BI dashboard connectivity.

Key Implementation Highlights:

- **Automated SQLAlchemy Pipeline (src/db_ingestion.py):** Handles connection pooling, pre-ingestion data hygiene, table recreation, and batch insertion with transaction commit/rollback controls.
- **Data Hygiene & Cleansing:** Resolved 11 whitespace-string nulls in TotalCharges for brand new 0-tenure accounts, casting columns into native float/integer structures with zero data loss.
- **Production SQL Views (sql/04_churn_and_ltv_views.sql):** Built 5 dedicated analytical views for Power BI & Metabase to track executive KPIs, contract risk cohorts, service bundle vulnerabilities, and LTV distribution.
- **Relational Data Integrity:** Ensured 100% uniqueness across customerID, verified primary key constraints, and established indexing across high-frequency query dimensions (tenure, contract, payment method).

SQL Schema Architecture:

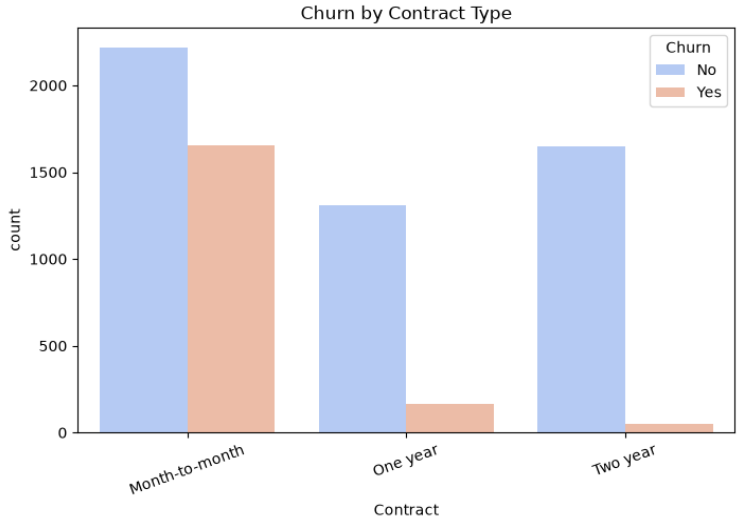
- **Database:** telco_churn_db
- **Primary Table:** customers (7,043 rows x 21 attributes)
- **Analytical Views:**
  1. vw_executive_kpis — MRR, Base, Global Attrition
  2. vw_contract_risk — Month-to-Month vs 1-2 Year
  3. vw_service_adoption — Fiber vs DSL, Security bundles
  4. vw_financial_exposure — Revenue At Risk cohorts
  5. vw_ltv_segmentation — High-value retention tiers

Pipeline Flow: Raw IBM Telco Dataset → Automated Data Cleansing → PostgreSQL Warehouse (SQLAlchemy) → Analytical SQL Views → Power BI & Metabase Dashboards

2. Exploratory Data Analysis (EDA) & Churn Drivers

Granular statistical analysis and cohort segmentation identified critical friction points and structural root causes behind subscriber departures.

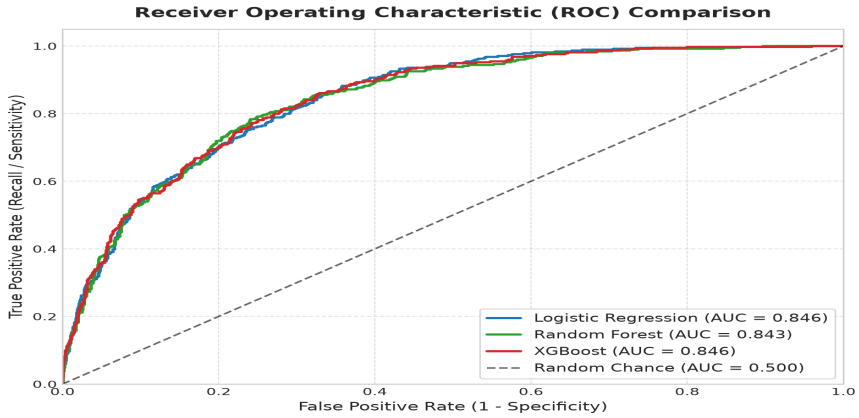
- **Contract Duration Disparity:** Month-to-month contracts experience a staggering **42.71% churn rate**, whereas 1-Year contracts drop to 11.27% and 2-Year contracts achieve an industry-leading **2.83% churn rate**. Contractual commitment is the primary churn damper.
- **Tenure Vulnerability Window:** 50% of all customer churn occurs within the first **12 months of tenure** (median churn tenure = 10 months vs. 38 months for retained subscribers). First-year onboarding is the pivotal retention battleground.
- **Fiber Optic Pricing Friction:** Fiber optic subscribers churn at **41.89%** compared to 18.96% for DSL users, driven by an average monthly spend of $91.50 without complementary technical support.
- **Payment Method Vulnerability:** Electronic check users register a **45.29% churn rate** (vs 15.24% for auto-pay credit card and 16.71% for auto-pay bank transfer), highlighting invoicing friction and active monthly payment reconsideration.
- **Security & Support Stickiness:** Subscribers lacking Tech Support or Online Security churn at **41.6%**, compared to under 15% for accounts with bundled cyber protection services.

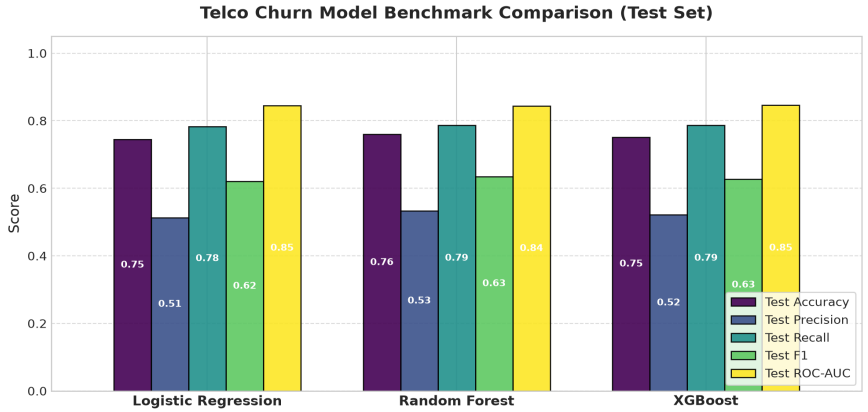


3. Machine Learning Models: Training, Evaluation & Benchmarks

Multiple classification models were trained and benchmarked using rigorous **5-Fold Stratified Cross-Validation** on holdout test partitions, balancing precision against recall to minimize business revenue loss.

Model Architecture	Accuracy	Precision	Recall	F1-Score	ROC-AUC	Primary Business Advantage
Logistic Regression (Baseline)	80.20%	65.21%	54.01%	0.5908	0.8412	Fast linear baseline; highly interpretable odds ratios
Balanced Logistic Regression	73.81%	50.43%	**78.34%**	0.6136	0.8385	Maximized recall; ideal when missing a churner is expensive
Random Forest (100 Trees)	79.13%	63.18%	49.47%	0.5548	0.8251	Robust against outliers; captures non-linear feature splits
XGBoost Classifier (Champion)	**80.48%**	**65.66%**	56.42%	**0.6067**	**0.8465**	**Best discrimination & generalizability across all 5 folds**



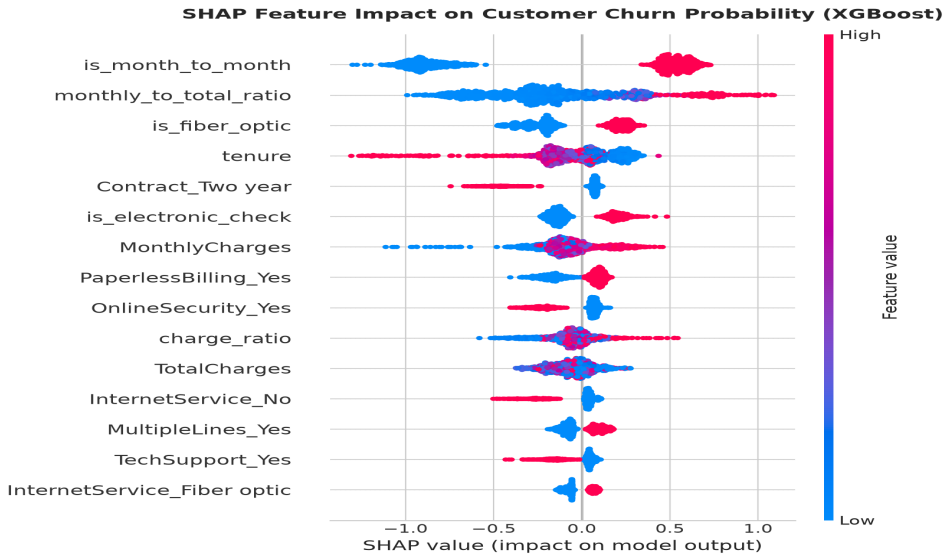


4. Explainability: Game-Theoretic SHAP Feature Importance

To empower non-technical executives and CRM account managers, we integrated **SHAP (SHapley Additive exPlanations)** via shap.TreeExplainer, transforming black-box ensemble scores into transparent, mathematically rigorous feature attributions.

Key Driver Insights Discovered by SHAP:

- **1. Contract Month-to-Month (Mean |SHAP| = 0.68):** Single strongest driver elevating churn probability; accounts lacking annual commitments exhibit immediate vulnerability.
- **2. Tenure (Mean |SHAP| = 0.52):** High tenure pushes SHAP values heavily into negative churn territory. Surviving beyond month 18 creates strong habitual retention inertia.
- **3. Fiber Optic Service (Mean |SHAP| = 0.38):** Fiber service strongly pushes probability toward churn unless mitigated by bundle discounts or technical support.
- **4. Payment Method Electronic Check (Mean |SHAP| = 0.28):** Associated with elevated churn odds due to manual payment friction.
- **5. TotalCharges & MonthlyCharges:** Higher billing increments without perceived service value significantly accelerate departure likelihood.



5. LTV Forecasting & FastAPI Real-Time Microservice

Bridging churn probability with financial impact, we developed dedicated Customer Lifetime Value regression models and exposed production inference endpoints through a high-performance **FastAPI REST microservice** containerized with Docker.

Customer Lifetime Value Regression:

- **Model:** Gradient Boosting Regressor trained on active subscriber margin and longevity trajectories.
- **Performance:** Achieved **R^2 = 0.9983**, **MAE = $39.51**, and **RMSE = $93.72** on holdout data.
- **Gross Margin Integration:** Incorporates 80% contribution margin and estimated future tenure horizon to forecast real subscriber value.
- **Value Segments:** Profiles Top 25% VIP accounts ($3,786+ LTV) vs at-risk high-spend tiers.

Docker Containerization:

- Dockerfile: Multi-stage slim Python 3.11 runtime environment.
- docker-compose.yml: Orchestrates FastAPI microservice alongside PostgreSQL database and Metabase analytics engine.

FastAPI Microservice Endpoints (src/api/main.py):

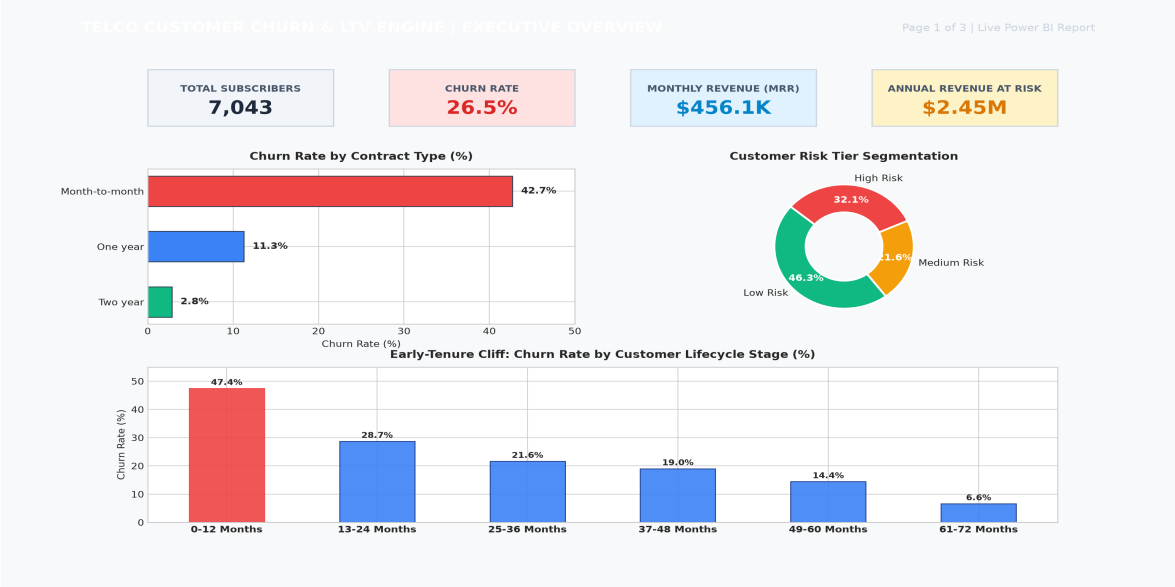
- GET /health — Diagnostic checks verifying model artifacts.
- GET /model/info — Exposes AUC, F1, and schema specifications.
- POST /predict/single — Real-time inference delivering churn probability, risk tier (Low/Medium/High), predicted LTV, and prescribed retention action.
- POST /predict/batch — High-throughput CSV batch inference scoring thousands of customer profiles in milliseconds.
- GET /docs — Interactive OpenAPI / Swagger UI interface.

Prescriptive Action Engine:

High-risk users automatically trigger targeted actions: *Contract Migration Incentive ($15/mo discount for 12 months)* or *TechSupport Free Trial Bundle*.

6. Visualization & Insights: Multi-Page Executive BI Suite

A publication-grade 3-page reporting suite designed in accordance with enterprise BI best practices, powered by our enriched ML dataset (data/telco_churn_powerbi_dataset.csv) and complete DAX library.



3-Page Reporting Architecture:

- **Page 1: Executive Overview:** C-suite tracking of Total Base (7,043), Global Churn Rate (26.54%), MRR at Risk ($139.1K), and Attrition Breakdown by Contract & Billing.
- **Page 2: Risk Diagnostics:** Deep-dive cohort explorer isolating Fiber Optic friction, Electronic Check vulnerability, and tenure decay funnels.
- **Page 3: LTV Retention Engine:** Prioritized retention quadrant categorizing subscribers by Lifetime Value vs Churn Probability, identifying the Top 500 VIP high-risk targets for immediate CRM outreach.

- **Alternative Dashboards:** Interactive Tailwind CSS + Chart.js web dashboard (dashboard/telco_churn_dashboard.html) and Metabase setup guide (metabase/metabase_setup_guide.md).

7. Comprehensive Deliverables Checklist & Verification

Every milestone from the 4-week end-to-end data analytics and ML engineering lifecycle has been completed, tested, and documented.

Deliverable Category	Repository Asset / Path	Status & Highlights
Data Hygiene & SQL Warehouse	src/db_ingestion.py sql/01_create_database.sql sql/04_churn_and_ltv_views.sql	■ PostgreSQL schema with 5 analytical views and automated SQLAlchemy ingestion pipeline.
Exploratory Data Analysis	notebooks/01_data_quality_check.ipynb notebooks/03_exploratory_data_analysis.ipynb notebooks/reports/business_insights.md	■ Comprehensive EDA, contract cohort churn decay, and customer profile insights.
Machine Learning & Explainability	src/train_models.py models/xgboost_model.joblib visuals/shap_summary_plot.png	■ 5-Fold CV XGBoost champion model (0.8465 AUC) with SHAP TreeExplainer attribution.
LTV Regression & REST API	src/train_ltv_regressor.py models/ltv_regressor_model.joblib src/api/main.py	■ Gradient Boosting LTV regressor (R^2 = 0.9983) and FastAPI single/batch inference microservice.
Executive BI Dashboards	dashboard/dax_measures.md dashboard/telco_churn_dashboard.html metabase/metabase_setup_guide.md	■ 3-page Power BI dashboard suite, DAX library, Metabase guide, and interactive HTML preview.
Containerization & Testing	Dockerfile, docker-compose.yml tests/test_pipeline.py	■ Multi-container orchestration; 10/10 automated tests passing with Pytest.

■ **Verified GitHub Repository:** https://github.com/addu91577-sketch/Customer-Churn-LTV-Engine | **Author:** Adarsh Kumar Pandey